

Quang Anh Vu

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Education

Hanoi University of Science and Technology (HUST)

B.Sc. in Mechatronics Engineering

Hanoi, Vietnam

Aug. 2021 – Aug. 2025

- Cumulative GPA: 4.0/4.0, CPA: 3.4/4.0.
- Academic Excellence Scholarship, awarded for three academic terms.

Research Interests

Humanoid robot learning; whole-body locomotion and control; behavioral foundation models; safe and robust robot learning; sim-to-real transfer.

Publications

Manuscripts Under Review

1. **Quang-Anh Vu**, et al. “AR-FlowZ: Autoregressive Riemannian Latent Flows for Latent Skill Sequencing in Behavioral Foundation Models.” *Under review, IEEE International Conference on Robotics and Automation (ICRA), 2027*. [Project Page]
2. **Quang-Anh Vu**, et al. “Ukemi-SafeFall: Humanoid Safe Falling & Recovery.” *Under review, IEEE-RAS International Conference on Humanoid Robots (Humanoids), 2026*. [Project Page] [Demo]

Peer-Reviewed Publications

1. Anh-Dung Nguyen, Tien-Dat Vu, **Quang-Anh Vu**, and Thai-Viet Dang. “Research on Modeling and Object Tracking for Robot Arm Based on Deep Reinforcement Learning.” *MM Science Journal*, June 2025. [DOI]
2. Chi-Hung Nguyen, **Quang-Anh Vu**, Kim-Khoi Phung Cong, and Thai-Viet Dang. “Optimal Obstacle Avoidance Strategy Using Deep Reinforcement Learning Based on Stereo Camera.” *MM Science Journal*, October 2024. [DOI]

Research Experience

VinMotion

Robotics Researcher

Hanoi, Vietnam

May 2025 – Present

- Developed **AR-FlowZ**, an autoregressive Riemannian latent-flow method for sequencing skills from a frozen behavioral foundation model; distilled iterative flow inference into a **3.4 ms endpoint predictor**, yielding a **29× speedup** and improving multi-skill transition success from **75.9% to 78.8%**.
- Deployed AR-FlowZ motions on a physical **Digit humanoid** through a **10-link Cartesian tracking interface**, achieving mean joint tracking errors of **3.4° for the arms** and **11.9° for the legs** without frame-by-frame inverse kinematics.
- Developed a PPO whole-body protective-falling and recovery policy for **Ukemi-SafeFall** using **4,096 parallel MuJoCo environments** and Score-Matching Motion Priors.
- Deployed the SafeFall policy to physical humanoid hardware through a **C++/ONNX Runtime controller at 50 Hz**, achieving **98% recovery success** and a **1.04 s mean stand-up time** in physical disturbance tests.
- Deployed and tuned **deep-RL locomotion and motion-imitation policies** on a **1.7 m full-size humanoid**, accounting for backlash, friction, actuator torque limits, and hardware tracking errors. [Demo]
- Implemented sim-to-real transfer through **system identification, actuator dynamics modeling, observation-delay modeling, domain randomization**, and randomized external-force disturbances.
- Integrated a **language-driven humanoid pipeline** connecting natural-language commands, **ARDY, GMR motion retargeting, BFM-Zero execution**, and joint-level control on physical **Motion Mini** hardware.
- Built and deployed humanoid control pipelines connecting high-level task policies to control at 50hz on physical robots.

- Integrated a **ResNet18–U-Net visual localization model** with **PPO control** for robotic-arm pick-and-place experiments and calibrated the manipulator and sensing setup for physical testing.
- Built a **ROS 2 navigation pipeline** combining stereo vision, **YOLOv8 object detection** and **Deep Q-Network control** for mobile-robot obstacle avoidance.
- Conducted simulation and hardware experiments contributing to **two peer-reviewed publications** on reinforcement-learning-based manipulation and mobile-robot navigation.

Technical Skills

Robot Learning & Control: Reinforcement Learning (PPO, DQN), Motion Imitation, Behavioral Foundation Models, Whole-Body Control, Model Predictive Control, PD/Torque Control, Motion Retargeting, Teleoperation, Sim-to-Real Transfer

Simulation & Robotics Frameworks: MuJoCo, Isaac Lab, mjlab, PyBullet, ROS 2, ONNX Runtime

Sim-to-Real & System Modeling: System Identification, Actuator Modeling, Domain Randomization, Observation-Delay Modeling, External Disturbance Randomization

Programming & Development Tools: Python, C++, Git, CMake, Linux, Linux RT-PREEMPT

Robot & Hardware Platforms: Full-Size Humanoids, Digit, Motion Mini, Robotic Manipulators, OptiTrack, Motion-Capture Suits

Research Systems: FB-CPR, BFM-Zero, ARDY, GMR, AR-FlowZ

References

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